Enterprise Financial Formula Auditor & Dynamic Error Repair Engine

Block 2 Technical Architecture & Comprehensive Operational Documentation

1. Executive Summary & Operational Scope

Modern enterprise financial systems demand uncompromised data integrity. Manual transactional entry, unvalidated ERP feeds, and rigid formula dependencies frequently introduce severe calculation failures such as `#N/A`, `#DIV/0!`, `#VALUE!`, and text-formatted currency anomalies. This module establishes a robust, 4-stage operational pipeline that ingests raw transactional feeds, autonomously traps execution faults, sanitizes type mismatches, and reconciles net extended costs against subledger control accounts within a strict $0.05 tolerance threshold.

Corrupted General Ledger
Ingests a 100-row raw transactional dataset containing deliberate corporate error signatures.

Formula Repair Engine
Applies advanced error-trapping and coercion formulas to sanitize data streams automatically.

2. Tab 1: Corrupted General Ledger (01_Corrupted_General_Ledger)

The raw ingestion layer houses a 100-row transactional dataset engineered to simulate unverified ERP feeds with realistic corporate error signatures and formatting noise:

Anomaly Type	Field Affected	Corrupted Representation	Business Impact
#N/A	Vendor_ID	`VEND-999` / `#N/A`	Breaks master catalog lookups and halts subledger aggregation.
#DIV/0!	Raw_Extended_Cost	Quantity = `0`	Causes catastrophic division-by-zero execution halts.
#VALUE!	Quantity_Units	`INVALID_QTY` / text strings	Triggers data type mismatches during arithmetic multiplication.
Formatting	Unit_Price_USD	`" $15.50 "` / `"$88.00"`	Stored as text strings with leading spaces and currency symbols.

3. Tab 2: Formula Repair Engine (02_Formula_Repair_Engine)

The repair engine acts as the computational sanitization core. It deploys robust wrapper functions (`IFERROR`, `IFNA`, `VALUE`, `TRIM`, `SUBSTITUTE`) to intercept execution faults and coerce corrupted inputs into clean, calculable numerical formats without data loss.

Vendor ID Resolution Formula

To neutralize unmatched lookup failures and missing vendor codes (`#N/A`), the repair layer evaluates the raw vendor field and reroutes exceptions to a secure default control account:

```
=IF(OR('01_Corrupted_General_Ledger'!D2="VEND-999", '01_Corrupted_General_Ledger'!D2="#N/A", ISBLANK('01_Corrupted_General_Ledger'!D2)), "VEND-DEFAULT", '01_Corrupted_General_Ledger'!D2)
```

Quantity Sanitization & Coercion Formula

To eliminate `#VALUE!` errors caused by text strings in quantity columns, whitespace is stripped and string entries are coerced into integers, defaulting safely to `1` upon parsing failure:

```
=IFERROR(VALUE(TRIM('01_Corrupted_General_Ledger'!E2)), 1)
```

Currency & Price Cleaning Formula

Text-formatted currency strings containing embedded spaces and dollar signs are cleaned by stripping non-numeric formatting characters and converting the remainder into standard decimal numbers:

```
=IFERROR(VALUE(TRIM(SUBSTITUTE(SUBSTITUTE('01_Corrupted_General_Ledger'!F2, "$", ""), " ", ""))), 0)
```

Repaired Extended Cost & Audit Tagging

The clean extended cost is computed via safe multiplication (`Clean_Quantity × Clean_Unit_Price`), avoiding downstream `#DIV/0!` errors. Each row is dynamically tagged as `REPAIRED` or `VERIFIED CLEAN` for executive review.

4. Tab 3: Reconciliation Matrix (03_Reconciliation_Matrix)

The reconciliation matrix compares repaired general ledger extended costs against independent subledger control balances. It calculates absolute deviations and enforces compliance against an established tolerance threshold:

Metric Column	Cell Formula (Row 2 Reference)	Operational Function
Absolute Variance	`=ABS(B2 - C2)`	Computes non-directional financial deviation magnitude.
Tolerance Threshold	`0.05`	Establishes the maximum allowable rounding/reconciliation variance ($0.05).
Reconciliation Status	`=IF(D2 <= E2, "MATCHED", "VARIANCE DETECTED")`	Automates compliance auditing, tagging items within tolerance as `MATCHED`.

5. Tab 4: Audit Executive Console (04_Audit_Executive_Console)

The executive console consolidates system health metrics into a real-time dashboard summarizing ingestion volume, repair frequency, and balance reconciliation integrity:

Total Transactions Audited
`=COUNTA('01_Corrupted_General_Ledger'!A2:A101)` — Measures total ingestion throughput.

Errors Trapped & Repaired
`=COUNTIF('02_Formula_Repair_Engine'!H2:H101, "REPAIRED")` — Quantifies intervention volume.

Reconciliation Variances
`=COUNTIF('03_Reconciliation_Matrix'!F2:F101, "VARIANCE DETECTED")` — Flags unaligned balances.

System Health Score
`=(1 - (F4 / B4))` (Formatted as %) — Evaluates overall data pipeline stability.